

ZARI.ai - Exhaustive Project & Thesis Defense Manual

Step-by-Step Technical Guide: Architecture, EDA, EDL, SAM2, RAG & Hurdles Explained

1. Introduction, Project Vision & Real-World Problem Context

A. Background & Agricultural Impact in Pakistan

Agriculture forms the backbone of Pakistan's economy, with major crops like cotton, wheat, and sugarcane. However, the sector faces significant challenges, including water scarcity, soil degradation, and the impact of climate change. The introduction of modern agricultural practices and technologies, such as the use of Solanum tuberosum (potatoes) for food security and livelihoods. However, the introduction of modern agricultural practices and technologies, such as the use of Solanum tuberosum (potatoes) for food security and livelihoods. However, the introduction of modern agricultural practices and technologies, such as the use of Solanum tuberosum (potatoes) for food security and livelihoods.

B. Core Real-World Hurdles Addressed by ZARI.ai

1. Overuse and Misuse of Fertilizers and Pesticides: The excessive use of chemical inputs has led to soil degradation and water pollution. The active ingredients of these chemicals are often harmful to the environment and human health.

C. System Solution Overview

ZARI.ai is an end-to-end AI-powered system designed to address the core real-world hurdles in Pakistani agriculture. It integrates a robust architecture for data processing, analysis, and decision-making, leveraging advanced machine learning models and a comprehensive dataset pipeline.

2. Exploratory Data Analysis (EDA) & Dataset Pipeline Deconstruction

A. Raw Datasets vs. Master Curated V4 Dataset

The raw datasets consist of thousands of images collected from various locations across Pakistan, including urban areas, rural fields, and agricultural zones. These images are often of low quality, with significant noise and clutter, making them unsuitable for direct analysis.

* Raw Sample Count: ~68,000 uncleaned images with label clutter and background noise.

* Master V4 Curated Dataset: A highly refined and curated dataset of 10,000 high-quality images, meticulously labeled for accuracy.

B. Data Quality Audit & Deduplication Pipeline

1. Exact & Partial Duplication Detection: The pipeline uses advanced algorithms to identify and remove duplicate images, ensuring the dataset's integrity and reducing redundancy. This process involves comparing image hashes and metadata to detect exact and partial duplicates.

3. Step-by-Step System Architecture & Technical Workflow

Step 1: Image Acquisition & Preprocessing

Input images are first acquired from various sources, including field cameras, drones, and mobile devices. These images are then preprocessed to enhance their quality, involving steps such as normalization (scaling pixel values to the range [0, 0.225]) and denoising.

Step 2: Model A Crop Router Classification

Model A (EfficientNet-B0) achieves 99.48% accuracy on Macro

Step 3: Model B Evidential Deep Learning (EDL) Disease Diagnosis

The crop-specific evidential probabilities are calculated for each class.

* Evidential Probability: $p_k = \alpha_k / S$

* Evidential Uncertainty: $1 - \sum p_k$

Step 4: Selective Classification & Risk Control (SCRC) Gate

If evidential uncertainty is high, the model returns 'insufficient data'.

Step 5: SAM2 Foliar Leaf Mask Segmentation

Meta SAM2 (HuggingFace) is used to segment the leaves, isolating the plant from the background.

Step 6: Grad-CAM Heatmap Localization & Severity Calculation

Grad-CAM maps are generated to localize the disease region. The intersection area is used for severity calculation.

Step 7: Pathogen Weather Context Lookup

Injects localized weather data (humidity >85%, average temperature 25-30°C) into the model.

Step 8: Multilingual ChromaDB Vector RAG Search

Queries ChromaDB for relevant Pashto and Urdu agricultural documents using MiniLM-L12-v2 embeddings.

Step 9: Integrated Pest Management (IPM) Advisory Synthesis

Assembles recommendations based on retrieved documents and local agricultural ingredients (Moringa, Neem).

4. Major Hurdles Encountered & Engineering Solutions

* Hurdle 1: Neural Network Overconfidence on Non-Plant Images.

Solution: Replaced non-plant images with plant-related images in the dataset.

* Hurdle 2: Background Soil and Weed Noise Distorting Visual Coverage.

Solution: Integrated background noise reduction module using median filtering.

* Hurdle 3: Vision Transformer (Swin-Tiny) Grad-CAM Hook Incompatibility.

Solution: Swapped Swin-Tiny for EfficientNetV2, which has a compatible Grad-CAM hook.

* Hurdle 4: Low Initial Pashto Language Retrieval Alignment (0.2484).

Solution: Enhanced retrieval by using a Pashto-English similarity matrix from a parallel corpus.

* Hurdle 5: Hallucinated Pesticide Dosages and Unregistered Products.

Solution: Implemented a validation layer against the official DPP Pakistan pesticide database.

5. Complete Empirical Results & Error Trajectories

A. End-to-End Real System CUDA Latency

* Vision Inferen

B. Per-Crop Model B Classifier Test Metrics

* Tomato (13 c

C. Knowledge Distillation Student Metrics

* Distilled Stud
Grad-CAM com

6. Saved Visualization Plots & Artifact References

* Model A Crop Router Curves: ml_pipeline/reports/figures/01_model_a_crop_r_outer_curves.png

* Model B Clas

7. Complete Repository Parameter Count

* Model A Crop Router: 7,705,221 (7.71 M)

* Model B Tom

8. Thesis Defense Expected Q&A Preparation

Q1: What is the primary novelty of ZARI.ai?

A: ZARI.ai u
background-is
12.05 ms real-

Q2: Why did you separate Model A (Crop Router) and Model B (Crop Classifiers)?

A: Hierarchica
allowing Mode

Q3: How does EDL handle out-of-distribution (OOD) images?

A: EDL output
evidence, driv

Q4: What embedder is used for multilingual search?

A: paraphrase
shared vector

Q5: Why retain EfficientNetV2-B2 over Swin-Tiny?

A: EfficientNet
output format (